

Week 13: Specification-driven programming

Functional Programming in Lean

Learning goals

- ▶ Fixed formal specification
- ▶ Generated implementation and proof
- ▶ Proof dependency audit

Core concepts

- ▶ `partitionBy`: stable order in both outputs
- ▶ `List.filter`: reference behavior
- ▶ Implementation and theorem statements as separate objects

Lean example

```
theorem partitionBy_yes
  (p :  $\alpha \rightarrow \text{Bool}$ ) (xs : List  $\alpha$ ) :
  (partitionBy p xs).1 = xs.filter p := by
  sorry
```

Lean details

- ▶ Intentional sorry in the displayed template
- ▶ sorryAx in an axiom report: incomplete proof
- ▶ Human review of the specification

Live coding

- ▶ Fixed theorem statements
- ▶ Candidate from an LLM
- ▶ Compilation and `#print axioms`

Check

Output order under a membership specification

Exercise

- ▶ Counterexample to $(xs, [])$
- ▶ Review of generated proof